



Modelling Financial Markets

Chapter 17: Bubbles and Crashes



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Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BSADF: Backward Sup Augmented Dickey–Fuller (statistic)
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ EM: Expectation-Maximisation (algorithm)
- ▣ FRED: Federal Reserve Economic Data
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ JLS: Johansen–Ledoit–Sornette (model)
- ▣ LPPL: Log-Periodic Power Law
- ▣ LPPLS: Log-Periodic Power Law Singularity (model)
- ▣ LR: Likelihood Ratio (test)
- ▣ MC: Monte Carlo (simulation)
- ▣ NVIDIA: NVIDIA Corporation (company name, US chip designer)
- ▣ OLS: Ordinary Least Squares
- ▣ PSY: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- ▣ PWY: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States

Chapter Route and Outcomes

Route of this chapter

- Question: prices sometimes rise far faster than any plausible change in fundamentals, then lose most of their value; can statistics detect such episodes **in real time**, and say **when** they end?
- Part I: what a bubble is, run-ups and crashes in the data, tests for explosive prices and their date-stamping
- Part II: the LPPLS model, regime switching, crypto and AI stocks, a case study, one pass of the AI-assisted research loop
- Technical detail sits in the Appendix at the end

After this chapter you can

- separate the fundamental value from a rational bubble and state what can be tested
- test for explosive prices with GSADF and date-stamp episodes with BSADF, with valid critical values
- fit LPPLS, judge the stability of its critical time and evaluate its alarms honestly
- describe booms and busts with a Markov-switching model and test the number of regimes

Three Historical Lessons



Hendrik Gerritsz Pot, *Flora's Wagon of Fools* (c. 1637), a satire of the tulip trade

Image: Hendrik Gerritsz Pot (c. 1637); public domain;
Wikimedia Commons

- **Tulips, 1636–1637:** contracts on rare bulbs traded at very high prices until the market stopped in February 1637; part of the path fits the fundamentals of a new luxury good [Garber \(1990\)](#)
 - ▶ Lesson 1: calling something a bubble requires a model of the fundamental value
- **South Sea and Mississippi, 1720:** shares rose several-fold within months, then collapsed; a financial innovation, easy credit and a story about future profits [Kindleberger & Aliber \(2005\)](#)
 - ▶ Lesson 2: bubbles come with credit and with high turnover [Scheinkman & Xiong \(2003\)](#)
- **1929:** a boom partly financed with borrowed money ended in the October crash; Kindleberger's sequence (after Minsky): displacement, boom, euphoria, distress, panic [Kindleberger & Aliber \(2005\)](#)
 - ▶ Lesson 3: we need tests that work on a single time series, in real time, before the panic

Data Used in This Chapter

Series	Frequency	Period	Use
S&P Composite, real price and dividend	monthly	1871–2026	price-dividend ratio
48 US industry portfolios	monthly	1926–2026	run-ups and crashes
Nasdaq 100, S&P 500	daily, monthly	1990–2026	dot-com, AI
BET, BET-FI	daily, monthly	1997–2026	Bucharest Stock Exchange
Bitcoin, Ethereum, Dogecoin	daily, weekly	2014–2026	crypto booms
NVIDIA, Cisco, Tesla, GameStop, Strategy (formerly MicroStrategy)	daily, monthly	1990–2026	single stocks

- Closing values up to 18 September 2026; stocks adjusted for dividends and splits; weekly and monthly series use the last price of the period
- Sources: shillerdata.com for the S&P Composite; [Kenneth French Data Library](https://kennethfrench.com) for industries

Fundamental Value

- No-arbitrage alone gives $P_t = \mathbb{E}_t[M_{t+1}(P_{t+1} + D_{t+1})]$, with M the stochastic discount factor and D the dividend
 - ▶ **Model assumption:** a constant required (expected) return r , so $P_t = \frac{\mathbb{E}_t[P_{t+1} + D_{t+1}]}{1 + r}$; $\mathbb{E}_t$: expectation under the physical measure, given the information at t
- Solving forward gives the **fundamental value**: $F_t = \sum_{i \geq 1} \frac{\mathbb{E}_t[D_{t+i}]}{(1 + r)^i}$
 - ▶ Gordon case, dividends growing at $g < r$: $F_t = D_{t+1}/(r - g)$
- But $P_t = F_t + B_t$ also solves the equation whenever $\mathbb{E}_t[B_{t+1}] = (1 + r) B_t$
- B_t is the **rational bubble**: investors pay for it because they expect to resell it at a higher price [Blanchard & Watson \(1982\)](#)

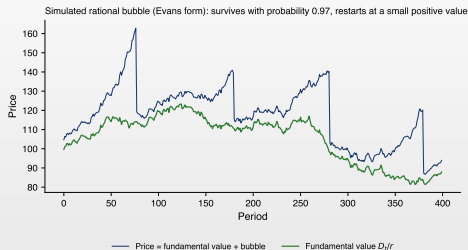
Why a Rational Bubble Must Explode

- From $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$: $\mathbb{E}_t[B_{t+h}] = (1+r)^h B_t$, an **explosive** path
 - ▶ The bubble grows faster than dividends ($r > g$), so its share of the price rises over time
- A negative bubble would eventually push the expected price below zero, and a bubble that is zero today must stay zero: rational bubbles are positive and exist from the first trade [Diba & Grossman \(1988\)](#)
- A bubble is the failure of the **transversality condition** $\lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$: the price contains a value never paid out as dividends (derivation in the Appendix)
- **Testable implication**: if dividends are at most $I(1)$, then without a bubble the price and the price-dividend ratio are **not explosive**
 - ▶ $I(1)$: integrated of order one; the first difference is $I(0)$ (stationary), while the level is not

Bubbles That Burst: Blanchard and Watson

- Each period the bubble survives with probability π and collapses with probability $1 - \pi$ [Blanchard & Watson \(1982\)](#)
 - ▶ $B_{t+1} = \frac{1+r}{\pi} B_t + \varepsilon_{t+1}$ with probability π , $B_{t+1} = \varepsilon_{t+1}$ otherwise
- Expected growth is still r ; while it survives the bubble grows faster, at $(1+r)/\pi - 1$, to pay for the crash risk
- Expected lifetime: $1/(1-\pi)$ periods, a geometric waiting time
- After a collapse $B = \varepsilon$ can be negative, which Diba–Grossman rule out: the simulation on the next slide restarts from a positive value (Evans form) [Evans \(1991\)](#)

A Simulated Rational Bubble



- Fundamental value D_t/r , random-walk dividends; $r = 1\%$, $\pi = 0.97$: growth close to 4.1% per period while alive and large, expected life 33.3 periods
- Evans form: collapse to $\delta = 5$, survival $B_{t+1} = \delta + \frac{1+r}{\pi}(B_t - \frac{\delta}{1+r})$ times noise of mean 1, so $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$ exactly
- In 400 periods the bubble grew large and burst 4 times; at its largest the price was 1.43 times the fundamental value
- Between episodes the price tracks fundamentals, so a full-sample test sees mostly “normal” data: the tests of this chapter use **windows**

Bubbles Beyond Rational Expectations



Robert J. Shiller, Nobel Prize 2013

Photo: Presidential Office Building, Taiwan (2017);

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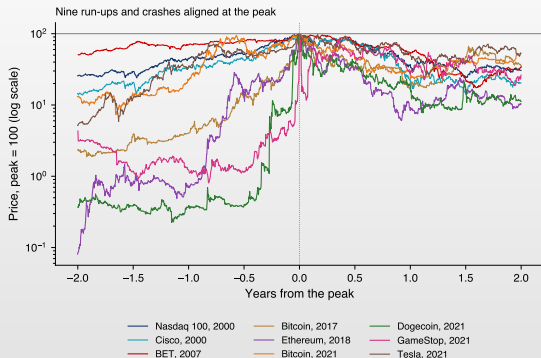
- **Feedback and narratives:** rising prices attract new buyers, “new era” stories spread by word of mouth [Shiller \(2015\)](#)
- **Disagreement and short-sale limits:** the price includes the option to resell to a more optimistic buyer [Scheinkman & Xiong \(2003\)](#)
- **Riding the bubble:** informed traders stay in as long as others do, since no one can burst it alone [Abreu & Brunnermeier \(2003\)](#), [Brunnermeier & Nagel \(2004\)](#)
- All three give prices that grow faster than fundamentals for a while: the explosive signature we will test

Nine Episodes: Run-Up and Crash

Episode	Peak	2-year run-up	Max. fall	Trough	Back at peak
Nasdaq 100, 2000	27 Mar 2000	+288%	−83%	Oct 2002	Nov 2015
Cisco, 2000	27 Mar 2000	+592%	−89%	Oct 2002	Aug 2021
BET, 2007	24 Jul 2007	+97%	−83%	Feb 2009	Mar 2021
Bitcoin, 2017	16 Dec 2017	+4186%	−83%	Dec 2018	Nov 2020
Ethereum, 2018	13 Jan 2018	+124004%	−94%	Dec 2018	Feb 2021
Bitcoin, 2021	8 Nov 2021	+667%	−77%	Nov 2022	Mar 2024
Dogecoin, 2021	7 May 2021	+26897%	−92%	Jun 2022	not yet
GameStop, 2021	27 Jan 2021	+2218%	−88%	Apr 2024	not yet
Tesla, 2021	4 Nov 2021	+1837%	−74%	Jan 2023	Dec 2024

- Peak: highest daily close in the episode; run-up: price change over the two years before the peak; maximum fall: to the lowest close within four years
- BET 2007: the only episode without a spectacular run-up in the last two years, yet with one of the deepest falls

Run-Ups and Crashes Aligned at the Peak



- Daily prices two years before and after each peak, peak = 100, log scale
- Two-year run-ups differ enormously, from +97% (BET) to +124004% (Ethereum); the falls are much more alike, 74–94%
- Common shape: an accelerating rise (convex on the log scale, the growth rate itself rises), a sharp top, a deep fall: the **explosive** behaviour the tests look for
- Selection bias: we picked these episodes because they crashed; many run-ups did not

The Dot-Com Bubble



The Pets.com sock puppet, mascot of an online pet-supply retailer that listed and closed in 2000

Photo: Atomic Taco (2013); CC BY-SA 2.0; Wikimedia

[Commons](#)

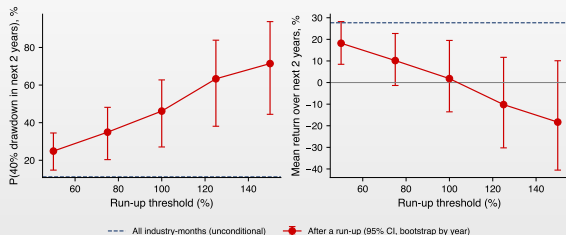
- After the peak of 27 Mar 2000: Nasdaq 100 -83% , Cisco -89% (table); Cisco regained its peak only in Aug 2021
- Hedge funds rode the bubble and cut positions before the fall [Brunnermeier & Nagel \(2004\)](#)
- Yet the internet did change the economy: a technology revolution can create both value and a bubble [Pástor & Veronesi \(2009\)](#)

Bubbles for Fama: The Design

- Fama's objection: a "bubble" is only a bubble if its end is predictable; otherwise it may be a rational reaction to news [Fama \(2014\)](#)
- Greenwood, Shleifer and You answer with **all** US industries since 1926, not with famous episodes only: do sharp run-ups predict crashes, or low returns? [Greenwood, Shleifer & You \(2019\)](#)
- **Run-up**: value-weighted industry return over the past 2 years above a threshold (e.g. 100%), both raw and net of the market
 - ▶ And a raw return of at least 50% over the past 5 years (from 1931), so that recoveries from slumps are not counted (Section 2 of the paper)
 - ▶ First month of the run-up; a new episode in the same industry only two years later
- **Crash**: a drawdown of at least 40% from a previous high within the next 2 years
- Our replication, same sample rules: the first 48 of the 49 industries of the [Kenneth French Data Library](#) (without "Other"), months with at least ten firms: 41 679 industry-months, 1926–2026

What Do Sharp Run-Ups Predict?

Bubbles for Fama, 48 US industries 1926-2026: two-year run-ups above the threshold, raw and net of the market

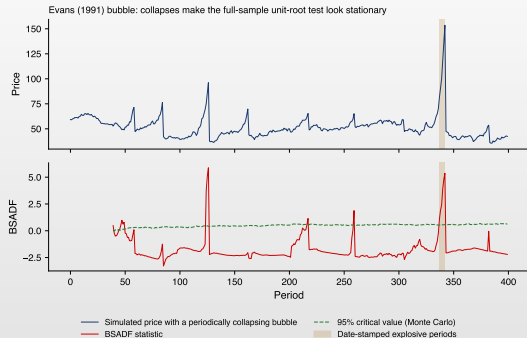


- After a run-up above 100% (52 episodes in 27 industries), a 40% crash followed in 46% of cases, 95% CI [27%, 63%], against 11% for a random industry-month; above 150%: 71% ($n = 21$)
 - CI: bootstrap over calendar years, because run-ups cluster in time (e.g. 1928–1929, 1967–1968, 1980); effective sample size in the Appendix
- The mean return over the next 2 years is not significantly different from zero: 2% after a 100% run-up, CI [–14%, 20%]: no statistically established negative mean return
- Run-ups raise the **risk** of a crash, not the predictable loss: price run-ups alone do not give a profitable timing signal Greenwood, Shleifer & You (2019)

Right-Tailed ADF and Collapsing Bubbles

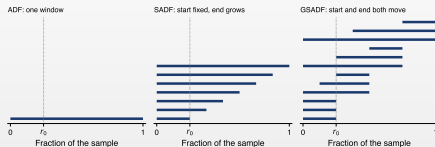
- Regression on the log price y_t : $\Delta y_t = a + b y_{t-1} + \varepsilon_t$; $ADF = \hat{b}/SE(\hat{b})$; here with a constant and no lagged Δy
- Bubble test: $H_0 : b = 0$ (unit root) vs $H_1 : b > 0$ (explosive root), the **right tail**; the left-tail Dickey–Fuller table does not apply [Phillips, Wu & Yu \(2011\)](#)
- Evans: a bubble that collapses periodically makes the **full-sample** series look stationary or like a random walk: low power; collapses give the full-sample slope a negative sign (Appendix) [Evans \(1991\)](#)
- Solution: run the ADF regression on many **windows** and take the largest statistic
 - ▶ SADF (Phillips, Wu and Yu): windows start at the first observation and grow [Phillips, Wu & Yu \(2011\)](#)
 - ▶ GSADF (Phillips, Shi and Yu): windows may start and end anywhere [Phillips, Shi & Yu \(2015a\)](#)

Evans Bubble: Full Sample vs Windows



- 400 simulated periods; BSADF (defined below) against its 95% critical value from 2000 random-walk paths
- Full-sample right-tailed ADF: -5.49 vs 95% critical value -0.09 : no rejection, the bubble is invisible
- SADF 0.98 (critical value 1.46) and GSADF 5.88 (critical value 2.21)
- BSADF crosses its critical value in several bursts, but only one lasts long enough to be date-stamped: the minimum-duration rule trades short bursts for fewer false alarms

Three Ways to Choose Windows: SADF, GSADF, BSADF

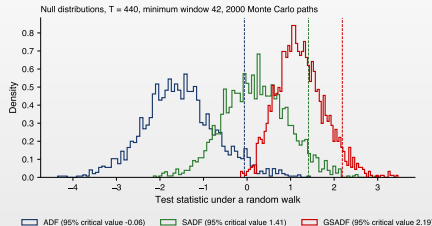


- $ADF_{r_1}^2$: the ADF statistic on the window from fraction r_1 to fraction r_2 of the sample; each bar in the chart is one ADF regression
- $SADF(r_0) = \sup_{r_2 \in [r_0, 1]} ADF_0^2$ Phillips, Wu & Yu (2011)
- $GSADF(r_0) = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^2$ Phillips, Shi & Yu (2015a)
- Backward SADF at the end point r_2 : $BSADF_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^2$; $GSADF = \sup_{r_2} BSADF_{r_2}$
 - $BSADF_{r_2}$ uses only data up to r_2 : it can be computed **in real time**
- Minimum window rule: $r_0 = 0.01 + 1.8/\sqrt{T}$, e.g. $T = 440$ months: $r_0 = 0.096$, 42 months Phillips, Shi & Yu (2015a)

Date-Stamping: When Did It Start and End?

- **Origination:** the first date at which BSADF exceeds its critical value [Phillips, Shi & Yu \(2015a\)](#)
 - ▶ The critical value sequence cv_{r_2} : 95% quantile of $BSADF_{r_2}$ under the null, date by date
- **Termination** (PSY, eq. 8): the first date after $\hat{r}_e + \delta \ln(T)/T$ at which BSADF falls below its critical value, δ depending on the frequency [Phillips, Shi & Yu \(2015a\)](#)
- **Rule used in this chapter:** the minimum-duration rule of Vasilopoulos, Pavlidis and Martínez-García (R package exuber): runs of consecutive exceedances shorter than L are dropped [Vasilopoulos, Pavlidis & Martínez-García \(2022\)](#)
 - ▶ $L = \lceil \ln T \rceil$ ($\delta = 1$); $T = 440$: 7 months. Start: first exceedance; end: last exceedance (the package dates the end one period later). Eq. 8 would instead extend a one-date blip to L dates
 - ▶ Real time: the start is dated in hindsight; the episode is **confirmed** only after L consecutive exceedances, $L - 1$ periods later
- PWY date-stamping uses the recursive ADF_0^2 instead: it may miss a second bubble after a collapse [Phillips, Shi & Yu \(2015b\)](#)

Null Distributions and Critical Values



- Random walk with a small drift, $y_t = T^{-1} + y_{t-1} + \varepsilon_t$, 2000 paths of $T = 440$; 95% critical values: a single ADF -0.06 (negative!), SADF 1.41, GSADF 2.19 [Phillips, Shi & Yu \(2015a\)](#)
- A supremum over more windows shifts the distribution to the right: the GSADF bar is the price of searching over about 80 000 windows; simulate it for each T , r_0 and lag order
- Asymptotics (Appendix): the null limit is pivotal; against mildly explosive roots the tests are consistent; date-stamping keeps a 5% false-alarm chance at every date [Phillips, Shi & Yu \(2015b\)](#)

Changing Volatility: Wild Bootstrap

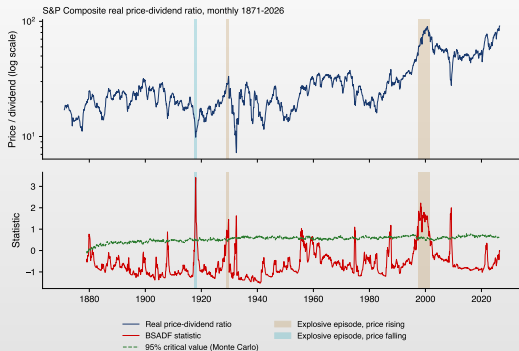
- ▣ Monte Carlo critical values assume shocks with constant variance; real prices have volatility clusters and shifts
- ▣ A burst of volatility can mimic explosiveness and inflate the rejection rate [Harvey et al. \(2016\)](#)
- ▣ **Wild bootstrap**: multiply the observed, demeaned changes Δy_t by random signs $w_t = \pm 1$, rebuild the path, recompute BSADF [Phillips & Shi \(2020\)](#)
 - ▶ The bootstrap null keeps the volatility pattern of the data but has no explosive root
 - ▶ Centred with the full-sample mean and resampling the whole path: **retrospective** critical values; in real time, recompute them with data up to each date
- ▣ Validity: the bootstrap data have a unit root by construction and inherit the time pattern of $\text{Var}(\Delta y_t)$; demeaning removes the drift, as in the null model [Harvey et al. \(2016\)](#)

Size of GSADF When Volatility Changes: Monte Carlo

Null model, no bubble	GSADF, MC crit. value	GSADF, wild crit. value	False episode, MC	False episode, w
Constant volatility	4.1%	3.9%	49.1%	51.9%
Volatility $\times 3$ at $T/2$	32.0%	4.1%	63.8%	49.6%
Volatility $\times 1/3$ at $T/2$	4.3%	6.1%	48.0%	54.8%
GARCH(1,1), $\alpha = 0.10$, $\beta = 0.88$	16.6%	6.8%	58.1%	52.8%

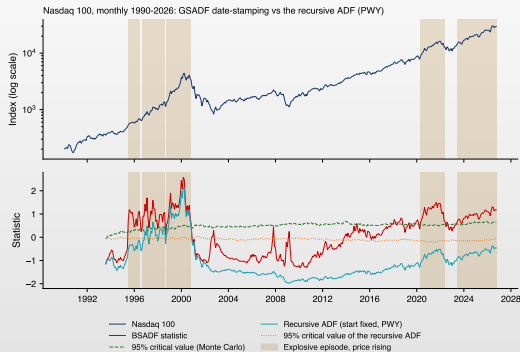
- Rejection rates at 5%; $y_t = T^{-1} + y_{t-1} + e_t$, $T = 400$, minimum window $w_0 = \lfloor r_0 T \rfloor = 40$ observations, 1 000 paths per row, each with its own wild bootstrap (199 replications); Monte Carlo error ± 1.4 points near 5%; false episode: BSADF above its pointwise 95% value for 6 periods
- A late rise in volatility is mistaken for explosiveness (32.0% with Monte Carlo, 4.1% with wild critical values); under GARCH the wild bootstrap removes most, not all, of the distortion; a fall in volatility is harmless [Harvey et al. \(2016\)](#)
- Even with constant volatility 49.1% of the paths contain a false date-stamped episode: the size is controlled date by date, not over the sample

S&P 500 Price-Dividend Ratio, 1871–2026



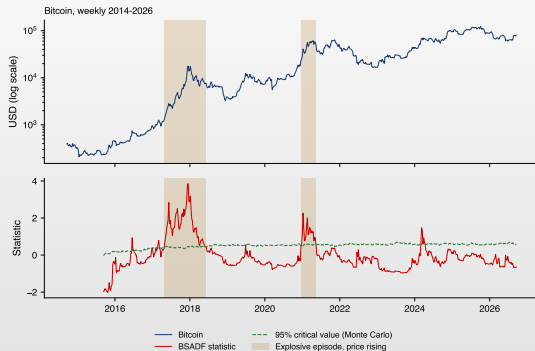
- Real price over real dividend, monthly, Jan 1871–Jun 2026, $T = 1865$, minimum window 96 months
- GSADF = 3.41 vs 95% critical value 2.39 (99%: 2.88); the full-sample ADF (-1.11) sees nothing
- Episodes: Aug 1917–Jun 1918; Nov 1928–Sep 1929; May 1997–Aug 2001
 - ▶ 1917–1918 is an accelerating **fall** of the ratio (real prices fell faster than real dividends): explosive, but not a bubble
- Dot-com: peak 90.2 in Aug 2000 vs a long-run mean of 29.8; the sample maximum, 91.2 in May 2026, lies in no episode: a high ratio is not an explosive one
- A ratio can explode through its denominator: always check both price and dividend

Nasdaq 100: GSADF vs PWY Date-Stamping



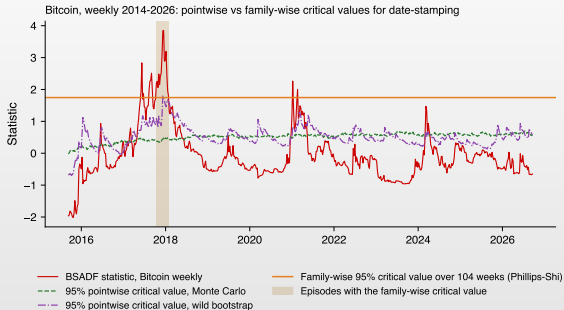
- Monthly log closes 1990–2026, $T = 440$, minimum window 42 months; 2 000 Monte Carlo paths
- GSADF = 2.57 (95% critical value 2.19); SADF = 2.14 (critical value 1.41)
- BSADF episodes: Jun 1995–Jun 1996; Aug 1996–Jul 1998; Sep 1998–Oct 2000; Apr 2020–May 2022; May 2023–ongoing
 - ▶ PWY (fixed start) episodes: Apr 1997–Nov 1997; Jan 1998–Jul 1998; Sep 1998–Oct 2000
- The recursive ADF starts in 1990 and is dragged down by the long flat period after 2000: it misses the post-2020 episodes that BSADF finds
- Dot-com episode dated from Jun 1995, 57 months before the peak of Mar 2000; confirmed in real time in Dec 1995

Bitcoin: Explosive Episodes



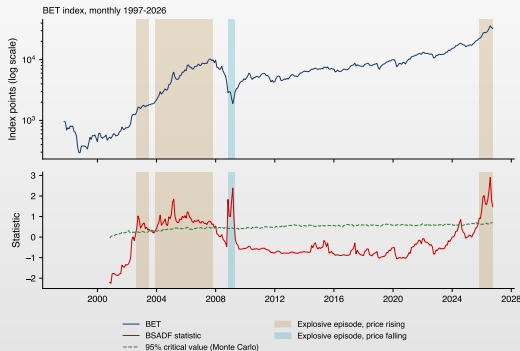
- Weekly log closes 2014–2026, $T = 626$, minimum window 51 weeks; Monte Carlo critical values
- GSADF = 3.85: Monte Carlo 95% critical value 2.27, wild bootstrap 2.82: rejected either way
- Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
 - ▶ Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- 2017: episode dated from Apr 2017, 8 months before the peak of 16 Dec 2017, confirmed in real time in Jun 2017; it ended 6 months after the peak
- The 2024–2025 rise to new highs is **not** flagged: steady growth is not the same as explosive acceleration

Date-Stamping Is a Multiple Test: Bitcoin



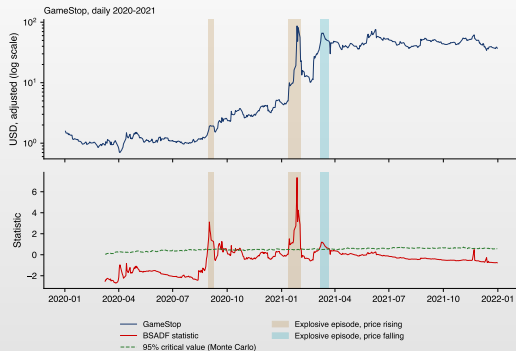
- BSADF is compared with its 95% critical value at each of 626 weeks: hundreds of dependent tests
- Family-wise error rate (FWER): the probability of at least one false alarm over the whole monitoring window; in 2 000 random walks with the Bitcoin T and w_0 : a crossing in 95%, an “episode” in 59% of paths
- Phillips and Shi: bootstrap the **maximum** of BSADF over the last τ_b monitoring dates (here $\tau_b = 104$ weeks); family-wise critical value 1.75, against pointwise values of at most 0.71 [Phillips & Shi \(2020\)](#)
- Episodes above the higher threshold: Oct 2017–Jan 2018; the 2020–2021 episode found with pointwise values does not survive
- A sensitivity check, not proved FWER control (design in the Appendix)

BET: The 2007 Boom and 2025



- Monthly log closes of the official BET, 1997–2026, $T = 348$, minimum window 37 months
- $GSADF = 2.90$ vs 95% critical value 2.21; episodes: Aug 2002–Jun 2003; Nov 2003–Oct 2007; Oct 2008–Mar 2009; Oct 2025–ongoing
- 2003–2007: the run-up before and around Romania's accession to the European Union (January 2007); BET then fell 83% from its peak of 24 Jul 2007
- Oct 2008–Mar 2009 is an accelerating **fall**: explosive can mean down as well as up
- The current episode (from Oct 2025) is still open: explosive does not mean it will crash; it means the rise is accelerating

GameStop, January 2021



A GameStop store, 3 February 2021, a week after the peak

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia Commons

- Daily log closes 2020–2021, $T = 504$, window 45 days; GSADF = 7.34, critical value 2.21
- January episode dated from 13 Jan 2021, 14 days before the closing high of 27 Jan 2021; confirmed on 22 Jan 2021
- A surge of weeks: it needs daily data; the test does not identify the mechanism (short covering)

 MFM_ch17_bet_gamestop

Real Time vs Hindsight

Episode	Start / Peak	Confirmed	Lead (m.)	Lead, conf.	Rise after conf.	Fall after peak
Nasdaq 100, monthly	Jun 1995 / Mar 2000	Dec 1995	57	51	+716%	−76%
BET, monthly	Nov 2003 / Jul 2007	Apr 2004	44	39	+299%	−83%
Bitcoin 2017, weekly	Apr 2017 / Dec 2017	Jun 2017	8	6	+590%	−83%
Bitcoin 2021, weekly	Dec 2020 / Apr 2021	Feb 2021	4	2	+66%	−75%
GameStop, daily	13 Jan 2021 / 27 Jan 2021	22 Jan 2021	0.5	0.2	+435%	−88%

- Start: first exceedance of the chain of BSADF episodes that leads to the peak (hindsight); confirmed: after L consecutive exceedances, the first date known in real time; leads in months; rise: confirmation to peak; fall: to the lowest close within 2 years (definitions in the Appendix)
- Even at confirmation the tests **warn early**, but the lead ranges from 5 days to about 4.2 years, and selling at confirmation would have missed most of the rise: they do not time the crash [Phillips, Shi & Yu \(2015b\)](#)

Crashes as Critical Points



Didier Sornette, ETH Zurich

Photo: Didier Sornette (2012); CC BY-SA 3.0
(Germany); Wikimedia Commons

- Johansen, Ledoit and Sornette (JLS): imitation among traders makes the **hazard rate** of a crash rise as a critical time t_c approaches [Johansen, Ledoit & Sornette \(2000\)](#)
- Rational investors stay in only if the price grows fast enough to pay for the rising crash risk
- Result: **faster-than-exponential** growth, decorated by oscillations that speed up near t_c
- t_c is the most probable end of the bubble, not a certain crash date [Sornette \(2017\)](#)

The LPPLS Equation

- $\ln p(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi), \quad t < t_c$
- A : log price at t_c ; $B < 0$: the super-exponential trend for a rising bubble; $0 < m < 1$: power-law exponent
 - ▶ The growth rate of the power-law trend, $-Bm(t_c - t)^{m-1}$, rises without bound as $t \rightarrow t_c$
- C, ω, ϕ : amplitude, angular log-frequency and phase of the log-periodic oscillations
- Seven parameters: A, B, C, ϕ and the nonlinear t_c, m, ω
- LPPL: the log-periodic power law pattern; LPPLS: the model with the finite-time singularity at t_c

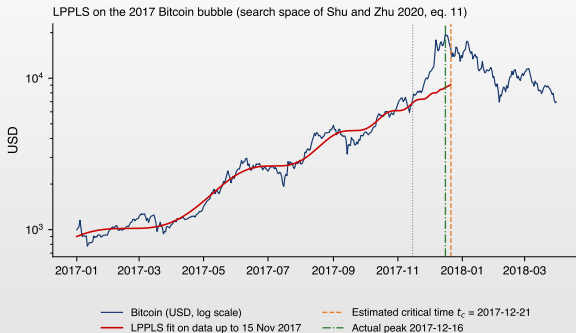
Calibration: Filimonov and Sornette

- Rewrite the cosine: $C \cos(\omega \ln(t_c - t) - \phi) = C_1 \cos(\omega \ln(t_c - t)) + C_2 \sin(\omega \ln(t_c - t))$ Filimonov & Sornette (2013)
 - ▶ A, B, C_1, C_2 are now **linear**: for given (t_c, m, ω) they come from OLS
- Search only over (t_c, m, ω) : a grid, then a Nelder–Mead refinement of the sum of squared residuals
- Search space of Shu and Zhu (Section 2.3, eq. 11): $m \in [0, 1]$, $\omega \in [1, 50]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/3]$, damping $\frac{m|B|}{\omega|C|} \geq 1$, an approximate non-negativity condition for the crash hazard rate, the crash probability per unit time given no crash yet (exact: $m|B| \geq |C|\sqrt{m^2 + \omega^2}$, Seminar 17, A8) Shu & Zhu (2020)
- Many local minima remain: different windows and starting points give different t_c Geraskin & Fantazzini (2013)

Qualified Fits: The Filter

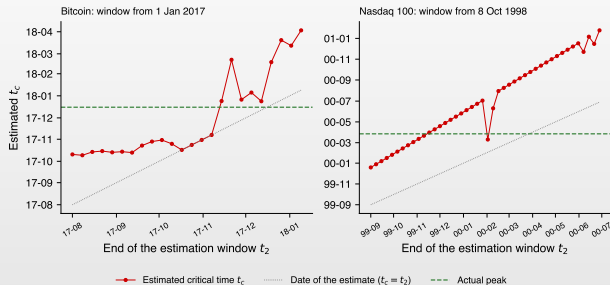
- A fit on $[t_1, t_2]$ is **qualified** if it passes all conditions of Shu and Zhu (Section 2.3, eq. 12), and $B < 0$ for a rising bubble Shu & Zhu (2020)
 - ▶ $m \in [0.01, 0.99]$, $\omega \in [2, 25]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/5]$
 - ▶ Number of oscillations (half-periods) $O = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$, so the oscillations are not noise
 - ▶ Largest relative error of the fitted price $\max_t |\hat{p}_t - p_t|/p_t \leq 0.15$
 - ▶ Lomb periodogram of the detrended residual $(t_c - t)^{-m}(\ln p_t - A - B(t_c - t)^m)$: log-periodicity significant at 10%
 - ▶ Residual unit-root check for $\ln \hat{p}_t - \ln p_t$: Dickey–Fuller and Phillips–Perron reject a unit root at 10% (a stationary, mean-reverting residual; this alone does not make it AR(1))
- $[t_1, t_2]$: the estimation window; the filter is applied to each window separately

LPPLS on Bitcoin, 2017



- Fit on daily log prices from 1 January to 15 November 2017; the path after 15 November is out of sample
- The fit follows the 2017 rise and its oscillations closely: $m = 0.95$, $\omega = 10.54$, $B = -2.62$
- Estimated t_c : 21 Dec 2017; actual peak: 16 Dec 2017, i.e. 5 days earlier
- The fit fails the condition on relative error ≤ 0.15 : it does **not** pass the filter, even though it looks convincing

How Stable Is the Critical Time?

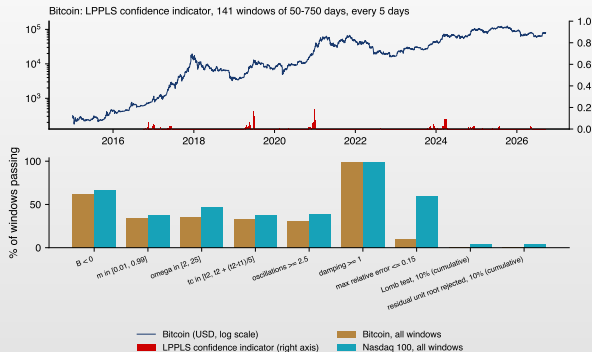


- A good in-sample fit is not a forecast: each point is one fit on a window with a fixed start, ending on the date on the horizontal axis (every 7 days)
- Bitcoin: before the peak (20 fits) t_c ranged from Oct 2017 to Feb 2018; mean absolute error 48 days; within 30 days of the peak: 20% of the fits
- Nasdaq 100: before the peak (30 fits) t_c ranged from Dec 1999 to Sep 2000; mean absolute error 70 days; $t_c \geq t_2$ is a constraint of the search, not a forecast
- Qualified fits before the peak: 0% (Bitcoin) and 0% (Nasdaq 100); the birth of a bubble is easier to diagnose than its burst [Demos & Sornette \(2017\)](#)

The LPPLS Confidence Indicator

- At each date t_2 , fit many windows $[t_2 - L, t_2]$ and report the **share of qualified fits** [Demirer et al. \(2019\)](#)
 - ▶ Grid of Shu and Zhu (Section 2.3): 141 windows, L from 750 down to 50 observations in steps of 5; t_2 every 5 observations; data up to t_2 only [Shu & Zhu \(2020\)](#)
 - ▶ Daily closes; Bitcoin trades every day, so one observation is one calendar day
- A high share means the bubble signature is robust to the choice of window
- Honest evaluation: is a signal followed by a fall of at least 20% within 90 days more often than no signal?
 - ▶ Fall measured from the price on the signal date; only dates whose 90 calendar days are fully observed (the last three months of data are dropped)

Bitcoin: Real-Time Confidence Indicator



- Top: share of the 141 windows with a qualified fit, every 5 days, 2016–2026; bottom: share of windows passing each condition (Lomb and residual tests cumulative)
- Bitcoin: the indicator was positive on 14% of the 2017 dates, 3% in 2021 and 12% in other years; maximum 0.18 in Dec 2020
- After a signal, a 20% fall within 90 days followed in 22% of cases, vs 37% without a signal (709 dates)
 - Nasdaq 100, 1997–2026: 4% after a signal vs 11% without (1482 dates)
- The raw difference is not yet evidence: signals come in runs and the 90-day outcome windows overlap (test on the next slide)

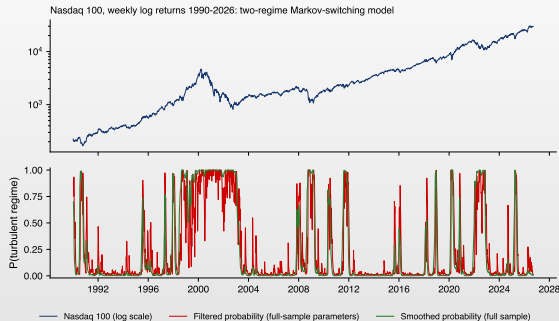
Is the Indicator Better than Chance? A Test

- $H_0: P(\text{fall} \geq 20\% \text{ within 90 days} \mid \text{signal}) = P(\cdot \mid \text{no signal})$; statistic $\Delta = \hat{\rho}_{\text{on}} - \hat{\rho}_{\text{off}}$
- Dates 5 days apart share up to 85 of their 90 outcome days: the two-sample z-test, which assumes independent dates, understates the standard error
- Circular block bootstrap of the (signal, outcome) pairs, blocks of 90 days, 9 999 replications; linear probability regression with Newey–West standard errors
 - ▶ Bitcoin: $\Delta = -15.5$ points, 95% CI $[-30.0, 1.8]$; block SE 8.1 vs independent-data SE 5.0; regression $p = 0.049$
 - ▶ The same test for the Nasdaq 100, with a power calculation: Seminar 17, Exercise B6
- The block-bootstrap CI contains zero, while the Newey–West regression is borderline ($p = 0.049$): weak evidence, and the point estimate is negative, i.e. if anything a lower crash rate after signals; a signal is **not** followed by a large fall more often than no signal, because signals come in steady rises, not just before crashes

The Markov-Switching Model

- Weekly log return $r_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t$, $\varepsilon_t \sim N(0, 1)$, regime $s_t \in \{1, 2\}$ unobserved [Hamilton \(1989\)](#)
 - ▶ Each regime has its own mean μ and volatility σ : a calm and a turbulent regime
- Regimes follow a Markov chain: $p_{ij} = P(s_t = j \mid s_{t-1} = i)$; expected duration of regime i : $1/(1 - p_{ii})$
- Estimation by maximum likelihood with the Hamilton filter, which updates $P(s_t = j \mid r_1, \dots, r_t)$ week by week
- **Filtered** probability: returns up to t for given parameters; **smoothed**: the whole sample (hindsight)
 - ▶ Our parameters are estimated on the full sample, so our filtered probabilities are only **pseudo real time**; genuine real time needs parameters re-estimated with data up to t
 - ▶ A worked filter step is in Seminar 17, Exercise A5

Nasdaq 100: Two Regimes

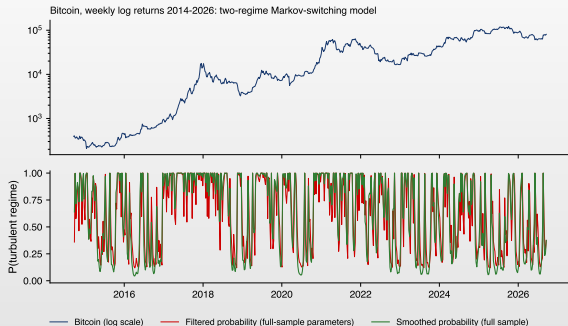


Regime	$\hat{\mu}$ (SE)	$\hat{\sigma}$	$\hat{\rho}_{ii}$	Dur.
Calm	0.43 (0.06)	2.30	0.987	75.4
Turbulent	-0.32 (0.27)	5.40	0.957	23.5

- Weekly log returns in %, 1990–2026, 1 915 weeks; duration in weeks; calm regime 76% of the weeks
- Turbulent: volatility more than twice as high, negative (insignificant) mean
- The turbulent spell around the 2000 peak began in Jun 1998 (smoothed probability above 0.5); filtered probability 0.99 at the end of March 2000: prices kept rising while volatility jumped

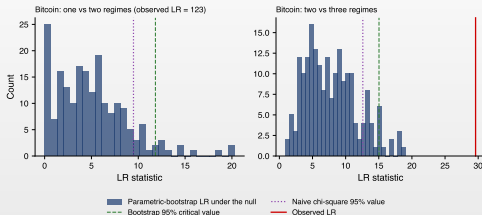
 MFM_ch17_markov_switching

Bitcoin: Two Regimes



- Weekly log returns 2014–2026, 626 weeks
- Calm: $\hat{\mu} = 0.26\%$ per week, $\hat{\sigma} = 2.93\%$, duration 3.8 weeks
 - ▶ Turbulent: $\hat{\mu} = 1.20\%$ (SE 0.58), $\hat{\sigma} = 11.22\%$, duration 6.3 weeks, 62% of the weeks
- Unlike equities, the turbulent regime has the **higher** mean; regimes are short: volatility states, not bubble phases
- Regimes ignore the price level (a slow, calm bubble stays calm): BSADF says “the price is accelerating”, the regime says “risk has changed”

How Many Regimes? A Bootstrap LR Test

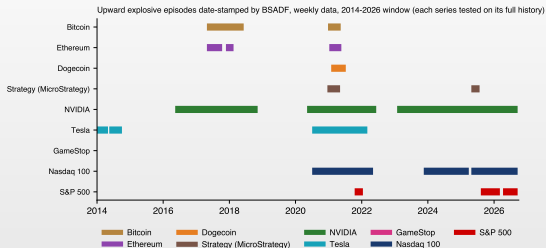


- H_0 : one regime (iid Normal returns) vs H_1 : two Markov regimes, $LR = 2(\hat{\ell}_2 - \hat{\ell}_1)$; not χ^2 : p_{11}, p_{22} are not identified under H_0 (Appendix [Hansen \(1992\)](#))
- **Parametric bootstrap**: simulate from the fitted H_0 model, refit both models, recompute LR; 199 replications;
 $p = (1 + \#\{LR^* \geq LR\}) / (B + 1)$
- One vs two: Bitcoin LR = 123.2 (95% value 11.8, $p = 0.005$); Nasdaq 100 LR = 475.7 (11.1, $p = 0.005$)
- Two vs three, Bitcoin: LR = 29.6, bootstrap 95% value 15.0 vs naive χ^2_6 value 12.6; $p = 0.005$ (Seminar 17, B8)
- One regime is rejected for both series; a third Bitcoin regime is also supported. Caveat: the one-regime null is Normal, so fat tails alone can favour two regimes

Are Crypto-Assets Bubbles?

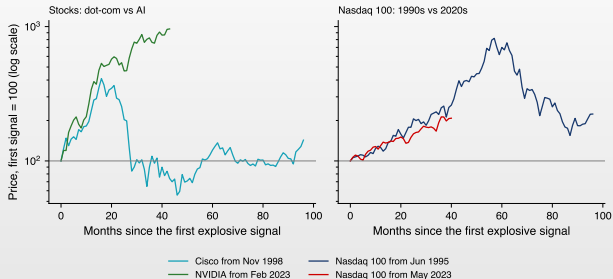
- No dividends: the fundamental value of Bitcoin is hard to define; one estimate puts it at zero [Cheah & Fry \(2015\)](#)
- Alternative: value from network adoption (Metcalf's law), with LPPL bubbles on top [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- Crypto-assets increasingly behave like an alternative asset class, not a pure bubble [Pele et al. \(2023\)](#)
- Without a fundamental value we can still test the price: is it explosive, and when?

Explosive Episodes in Crypto and Tech Stocks



- Weekly log closes; each series tested on its own history with its own Monte Carlo critical values; rising episodes only
- Crypto: Bitcoin Apr 2017–Jun 2018; Dec 2020–May 2021; Ethereum Apr 2017–Oct 2017; Nov 2017–Feb 2018; Jan 2021–May 2021; Dogecoin Jan 2021–Jul 2021: booms cluster in 2017 and late 2020–2021
- Strategy (formerly MicroStrategy; holds Bitcoin): Dec 2020–May 2021; Apr 2025–Jul 2025; Tesla: Nov 2013–May 2014; May 2014–Oct 2014; Jul 2020–Mar 2022; NVIDIA: May 2016–Nov 2018; May 2020–Jun 2022; Jan 2023–ongoing
- GameStop, weekly: the 3-week surge of January 2021 is shorter than the minimum duration; an accelerating **fall** also gives $b > 0$: always report the direction

Dot-Com vs AI



- Monthly closes, rebased to 100 at the first explosive signal of each episode (BSADF, monthly data)
- Cisco: first signal Nov 1998, peak after 16 months at 4.1 times the signal price; NVIDIA: first signal Feb 2023, 43 months later at 9.6 times, episode still open
- Nasdaq 100 from Jun 1995: peak after 57 months at 8.2 times; from May 2023: 2.1 times after 40 months, a slower rise than in the 1990s

Is There an AI Bubble? What the Tests Can Say

- ▣ Monthly BSADF flags NVIDIA (Mar 2016–Nov 2018; Feb 2020–May 2022; Feb 2023–ongoing) and the Nasdaq 100 (current episode from May 2023)
- ▣ The tests cannot separate a bubble from explosive fundamentals: fast-growing expected profits of a new technology can also make prices explosive [Pástor & Veronesi \(2009\)](#)
- ▣ History: Cisco's dividend-adjusted share price returned to its March 2000 level only in Aug 2021
- ▣ What we can say: the tests detect explosive price dynamics; the industry run-up evidence of Greenwood, Shleifer and You (2019, *Bubbles for Fama*) motivates watching crash risk but does not quantify it for these stocks, and the timing is unknown

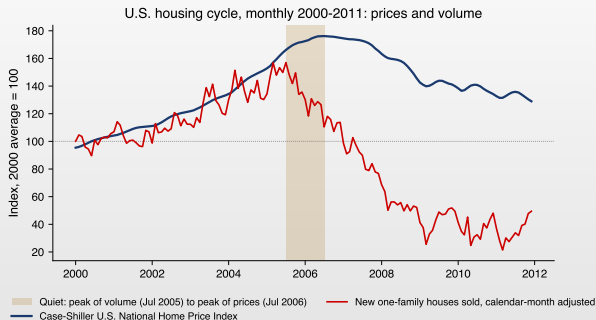
Case Study (1/5): the Research Question

- Paper: [DeFusco, Nathanson & Zwick \(2022\)](#), *Journal of Financial Economics*, 146(1), 205–229
 - ▶ Setting: the US housing cycle of 2000–2011, the boom and bust behind the 2008 crisis
- Question: who buys during a boom, and can speculation explain the joint path of prices, transaction volume and unsold listings (Introduction)?
 - ▶ Speculation, in the Harrison–Kreps definition the authors adopt (footnote 1): paying more for an asset because one has the right to resell it
- Three phases (Section 2 of the paper): the boom (prices and volume rise), the “quiet” (volume falls while prices still rise and unsold listings pile up), the bust
- Why it matters: the explosiveness tests of this chapter use prices only; Greenwood, Shleifer and You (2019, *Bubbles for Fama*) [Greenwood, Shleifer & You \(2019\)](#) also study turnover, issuance and volatility, and bubbles come with high turnover [Scheinkman & Xiong \(2003\)](#)
 - ▶ Quantities show who drives a boom, as blockchain flows did for the 2017 Bitcoin boom [Griffin & Shams \(2020\)](#)

Case Study (2/5): Data, Method and Headline Results

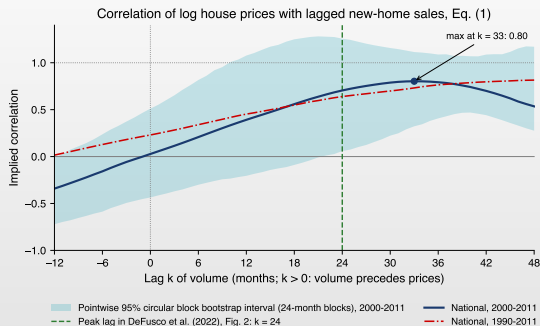
- Data (Section 1.1): 51 580 408 home sales from deeds records, 1995–2014, 115 metropolitan areas; **speculators**: short-term buyers who resell within three years, and non-occupant buyers (Section 1.2)
- Lead-lag (Eq. 1): $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$, $k = -12, \dots, 48$ months, p the demeaned log price, v the demeaned sales rate; implied correlation $\beta_k \text{sd}(v_{i,t-k}) / \text{sd}(p_{i,t})$
- Figure 2: the correlation peaks at a lag of 24 months; Figure 4: short-term sales explain 39% of the 2000–2005 rise in sales
- Table 2: one standard deviation more speculative volume goes with a price boom larger by 24.9 points and a bust deeper by 7.4 points: cities with more speculation had larger booms and larger busts
- Model (Sections 5–6, Table 6): extrapolating buyers with short horizons amplify the cycle; a 5% purchase tax cuts the bust from 8.2% to 1.1%

Case Study (3/5): Prices and Volume, 2000–2011



- FRED: Case-Shiller U.S. National Home Price Index and new one-family houses sold, 2000 average = 100; volume adjusted by calendar-month means, as in Figure 1 of [DeFusco, Nathanson & Zwick \(2022\)](#)
- Boom: sales peak at 157 in Jul 2005 (paper: 150% of the 2000 level); prices keep rising to 176 in Jul 2006
- Quiet: 12 months from the volume peak to the price peak, when sales are already down to 110 (unadjusted sales: 16 months)
- Bust: 2011 averages 35 for sales, 133 for prices; new-home sales exclude resales, so they cannot show who trades

Case Study (4/5): Volume Leads Prices



- Eq. (1) on the national series, without a constant; pointwise 95% band: circular block bootstrap, 24-month blocks, 2000 replications
- 2000–2011: maximum at $k = 33$ (0.80, CI [0.32, 1.11]); at the paper's $k = 24$ ours is 0.70, CI [0.06, 1.26]
- The band excludes zero for $k = 22, \dots, 48$, but one national cycle holds few independent blocks: wide intervals
- Direction as in the paper (volume first, by two to three years); its length is not precise (1990–2011: $k = 48$)



Case Study (5/5): Limits and an Open Question

- Data limit of our illustration: the public series are national and cover new homes only, so short-term and non-occupant volume cannot be built; Figures 1–2 can be illustrated, Table 2 cannot (Seminar 17, C3)
- Limits stated by the authors
 - ▶ Holding periods are realised, not intended (footnote 4; instrumental-variable check in the Online Appendix)
 - ▶ The non-occupant proxy misclassifies owners who give the property address for tax mail (Section 1.2)
 - ▶ The counterfactuals may overstate the role of short-term buyers, because selling horizons are exogenous in the model; credit and new construction are left out (Sections 4.3, 6.5.3, 7)
- Open question: does trading volume also lead prices in crypto booms, and is there a quiet (falling volume, rising prices) before the end of the Bitcoin episodes dated by BSADF?

An Open Question

Can bubble alarms be raised in real time with a controlled false-alarm rate, and do they carry crash information?

- BSADF date-stamping runs one right-tailed test per date: the size is controlled date by date, not over the whole monitoring period
 - ▶ Live research: monitoring rules for bubbles *and* crashes with controlled false alarms Whitehouse, Harvey & Leybourne (2023); real-time “econometric thermometers” for housing Shi & Phillips (2023)
- Formal version, for an alarm A_t and a crash $C_{t,h}$ (a fall of at least 20% within $h = 90$ days):
 $H_0: P(C_{t,h} | A_t = 1) = P(C_{t,h} | A_t = 0)$, H_1 : larger after an alarm
 - ▶ Falsified if the hit rate after alarms does not exceed the rate without alarms, under inference that respects overlapping windows
- Identification: explosive prices need not be a bubble
 - ▶ Explosive expected cash flows of a new technology leave the same signature Pástor & Veronesi (2009); run-ups raise crash risk without predictable average losses Greenwood, Shleifer & You (2019)

The Discovery Loop and Its Checks

- **Literature:** ask for papers on the false-alarm rate of sequential bubble monitoring; resolve every DOI on [Crossref](#) and find the table the answer quotes
- **Hypotheses:** rational bubble, explosive expected cash flows, falling discount rate, each with its own observable implication (e.g. an explosive price-to-earnings ratio vs explosive earnings)
- **Code:** replicate a known number first ($GSADF = 3.85$ for weekly Bitcoin), then extend
- **Robustness:** pre-register the grid of horizons and crash thresholds with a Holm correction; **critique:** ask the assistant to attack the result as a referee
- **Checks** the researcher must run (details and prompts in the Appendix)
 - ▶ No full-sample information in a date stamp; right-tail critical values; wild bootstrap under changing volatility; multiplicity over dates and assets; inference robust to overlapping outcome windows

Mini-Case: Does the LPPLS Alarm Predict Crashes?

- A typical assistant proposal: “a positive LPPLS confidence indicator announces a crash; compare hit rates with Fisher’s exact test”; alarms come in runs and outcomes overlap, so we test by circular rotations
- Bitcoin (709 dates, every 5 days): a fall of 20% within 90 days after 22% of alarms vs 37% without
 - ▶ Difference –15.5 percentage points; one-sided $p = 0.96$ for H_1 from 658 rotations of the alarm series
 - ▶ Nasdaq 100: 4% vs 11%, difference –7.1 points, $p = 0.98$
- Pre-registered grid of 4 horizons $\times$ 3 thresholds: smallest raw p is 0.76 (Bitcoin) and 0.69 (Nasdaq 100)
 - ▶ No cell rejects H_0 at 5%, before or after the Holm correction
- Interpretation: the indicator dates accelerating rises, but these tests do not support a higher probability of large falls after a signal (assumptions in the Appendix)

Your Turn: a Project Seed

- **Question:** can a monitoring rule for Bitcoin, the Nasdaq 100 and NVIDIA keep the probability of any false alarm at 5% over the whole period and still warn before large falls?
- **Replicate first:** GSADF = 3.85 for weekly Bitcoin, then the crash probability after run-ups above 100% in [Greenwood, Shleifer & You \(2019\)](#) (ours: 46%)
 - ▶ Extend: calibrate the alarm threshold by simulation over the whole monitoring period [Whitehouse, Harvey & Leybourne \(2023\)](#); evaluate the alarms with rotations or a block bootstrap
- **Deliverables:** a pre-registered grid fixed in a dated document before the evaluation, a notebook that regenerates every number, false-alarm rates and hit rates with valid p -values
- **AI log:** prompts that shaped the design in AI_USE.md; each wrong claim (left-tail critical values, a full-sample flag) and the test that caught it in AI_ERRORS.md ([templates](#))

Key Takeaways

- ▣ A bubble is price minus fundamental value; rational bubbles must be explosive, and bursting bubbles hide from full-sample tests
- ▣ GSADF tests for explosive episodes, BSADF date-stamps them in real time; with a late volatility rise Monte Carlo critical values reject 32.0% of true nulls, the wild bootstrap 4.1%; date-stamping needs family-wise control
- ▣ Across our episodes the tests warned months before the peak, but did not time the crash
- ▣ LPPLS describes the acceleration; its critical time moves with the window, and the confidence indicator is judged by a dependence-robust test of hit rates
- ▣ Sharp run-ups raise crash risk without a statistically established average loss: a risk-management signal, not a trading rule

Which Tool When?

Question	Tool
Has there been any explosive episode? When did it start and end?	GSADF with simulated critical values BSADF date-stamping, minimum duration $\lceil \ln T \rceil$; family-wise critical values for long monitoring
Volatility changes a lot (crypto, crises)	Wild bootstrap critical values
Is the rise accelerating super-exponentially?	LPPLS confidence indicator over many windows
Has the risk state changed?	Markov-switching filtered probabilities
How likely is a crash after a run-up?	Run-up conditioning across many assets (Greenwood, Shleifer and You 2019)

Check Yourself, and Next: Chapter 18

Retrieval questions

- Why does a full-sample right-tailed ADF miss a bubble that collapses periodically?
- Why is the 95% critical value of GSADF far above that of a single ADF?
- Why is a single LPPLS critical time not a crash forecast?
- Why must hit rates after alarms be tested with a block bootstrap or rotations?

Chapter 18: systemic risk

- When a bubble bursts, losses spread through banks and markets: contagion and systemic risk
- CoVaR (the value-at-risk, VaR, of the system when an institution is in distress) and SRISK (the capital shortfall of an institution in a crisis) measure how much one institution adds to the risk of the system
- Further reading: Franke, Härdle & Hafner (2019), Kindleberger & Aliber (2005), Shiller (2015), Sornette (2017)

Appendix: Transversality: Where the Bubble Term Comes From

- Substitute the pricing equation into itself h times: $P_t = \sum_{i=1}^h \frac{\mathbb{E}_t[D_{t+i}]}{(1+r)^i} + \frac{\mathbb{E}_t[P_{t+h}]}{(1+r)^h}$ (law of iterated expectations)
- Let $h \rightarrow \infty$: $P_t = F_t + B_t$ with $B_t = \lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h$
- **Transversality condition:** $\lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$; it holds $\Leftrightarrow B_t = 0 \Leftrightarrow P_t = F_t$
- Shift the limit one period: $\mathbb{E}_t[B_{t+1}] = \lim_h \mathbb{E}_t[P_{t+1+h}]/(1+r)^h = (1+r) B_t$, the martingale condition of the rational bubble
- Seminar 17, Exercise A1 derives the general solution and why $B_t \geq 0$ rules out a bubble that starts from zero [Diba & Grossman \(1988\)](#)

Appendix: Bubbles for Fama: Effective Sample Size

- ▣ The run-ups fall in 31 calendar years and overlap: consecutive 2-year windows of the same boom share months
- ▣ Year-block bootstrap SE (9.1 percentage points) vs binomial SE (6.9): a design effect of 1.7, i.e. about 30 independent observations
- ▣ Blocks of 2, 3 and 5 consecutive years (2-year windows of adjacent years overlap): $SE = 8.9, 8.7, 8.3$
- ▣ Unconditional mean 2-year return over all industry-months: 28%, above the CI of the mean return after a 100% run-up (−14% to 20%): returns after run-ups look lower than usual, but this is a descriptive comparison, not a formal test of the difference of two dependent estimates

Appendix: Why the Full-Sample ADF Fails: the Leverage of Collapses

- OLS slope: $\hat{b} = \sum_t \tilde{y}_{t-1} \Delta y_t / \sum_t \tilde{y}_{t-1}^2$, with $\tilde{y}_{t-1}$ the demeaned lagged log price
- Expansion: $\Delta y_t > 0$ while $\tilde{y}_{t-1}$ rises above its mean: positive terms
- Collapse: the most negative Δy_t occurs at the highest $\tilde{y}_{t-1}$: a few terms of the largest size and the opposite sign
- Evans form in logs, $y = \ln B$, collapse to $\delta = e^c$ with probability $1 - \pi$; for $B \gg \delta$ the survival log growth is $\approx \ln \frac{1+r}{\pi}$:
 $\mathbb{E}[\Delta y_t \mid y_{t-1}] \approx \pi \ln \frac{1+r}{\pi} + (1 - \pi)(c - y_{t-1})$, slope $-(1 - \pi) < 0$: apparent mean reversion although $\mathbb{E}_t B_{t+1} = (1 + r)B_t$
 Evans (1991)
- Windows that contain the expansion but not the collapse have $\hat{b} > 0$ when the log price accelerates there (next slide): a supremum over windows restores the power Phillips, Shi & Yu (2015b)

Appendix: Explosive in Levels, Accelerating in Logs

- A rational bubble is explosive in the **level**: $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$, an autoregressive root $1+r > 1$
 - ▶ In logs a pure bubble path $B_t = B_0(1+r)^t$ has constant growth: $\ln B_t = \ln B_0 + t \ln(1+r)$, a deterministic linear trend in logs, $b = 0$ in $\Delta y_t = a + b y_{t-1} + \varepsilon_t$
- The log price $\ln(F_t + B_t)$ **accelerates** while the bubble share grows: its growth rate rises from that of F towards r (a convex log path)
 - ▶ Once the bubble dominates, log growth is roughly constant again: the log-price regression then has little power
- So the log-price test targets a different alternative, **accelerating** log prices, not the level condition itself; the specification of the null and of the deterministic terms matters [Phillips, Shi & Yu \(2014\)](#)
- In this chapter we test log prices and log ratios (like PWY for the Nasdaq); PSY test the price-dividend ratio [Phillips, Wu & Yu \(2011\)](#), [Phillips, Shi & Yu \(2015a\)](#)

Appendix: Limit Theory (1): the Null Distribution

- Null model: $y_t = d T^{-\eta} + y_{t-1} + \varepsilon_t$, $\eta > 1/2$, ε_t iid $(0, \sigma^2)$: a random walk with an asymptotically negligible drift [Phillips, Shi & Yu \(2015a\)](#)
- By the functional central limit theorem and continuous mapping (PSY, Theorem 1, eq. 5), with $r_w = r_2 - r_1$ and W a standard Brownian motion:
 - $\text{ADF}_{r_1}^2 \Rightarrow \frac{\frac{1}{2} r_w [W(r_2)^2 - W(r_1)^2 - r_w] - \int_{r_1}^{r_2} W dr [W(r_2) - W(r_1)]}{r_w^{1/2} \left\{ r_w \int_{r_1}^{r_2} W^2 dr - \left(\int_{r_1}^{r_2} W dr \right)^2 \right\}^{1/2}}$
- σ and d disappear: the limit is pivotal. SADF and GSADF are suprema of the same functional, so their critical values depend only on r_0 and the deterministic terms
- $r_1 = 0$, $r_2 = 1$ gives the Dickey–Fuller limit with a constant: its 95% quantile is negative (simulated: -0.06), far from 1.645
- The drift and intercept specification changes the limit and the size [Phillips, Shi & Yu \(2014\)](#)

Appendix: Limit Theory (2): Mildly Explosive Alternatives

- Alternative: $y_t = \rho_T y_{t-1} + \varepsilon_t$ with $\rho_T = 1 + c/k_T$, $c > 0$, $k_T \rightarrow \infty$, $k_T = o(T)$; PSY use $k_T = T^\alpha$, $\alpha \in (0, 1)$
 - ▶ Closer to 1 than any fixed explosive root, farther than a local-to-unity root $1 + c/T$: the “mild” speed of real bubbles
- Phillips and Magdalinos: $\frac{\rho_T^T}{\rho_T^2 - 1} (\hat{\rho}_T - \rho_T) \Rightarrow \mathcal{C}$, a standard Cauchy law, without Gaussian errors [Phillips & Magdalinos \(2007\)](#)
 - ▶ Rate $k_T \rho_T^T$: exponentially fast; the ADF statistic diverges to $+\infty$
- Hence SADF and GSADF diverge too (for GSADF any window of at least $r_0 T$ observations inside the bubble suffices): the tests are consistent, power $\rightarrow 1$ [Phillips, Shi & Yu \(2015b\)](#)
- Finite samples: power depends on c , α and the duration; short bursts may stay below the bar, as in the Evans simulation

Appendix: Limit Theory (3): Consistent Date-Stamping

- Crossing times (PSY, eqs. 7–8): $\hat{r}_e = \inf\{r_2 \geq r_0 : \text{BSADF}_{r_2} > \text{scv}_{r_2}^{\beta_T}\}$,
 $\hat{r}_f = \inf\{r_2 \geq \hat{r}_e + \delta \ln(T)/T : \text{BSADF}_{r_2} < \text{scv}_{r_2}^{\beta_T}\}$
- $\text{scv}_{r_2}^{\beta_T}$: the $100(1 - \beta_T)\%$ quantile of BSADF on $\lfloor Tr_2 \rfloor$ observations
 - Consistency, $(\hat{r}_e, \hat{r}_f) \rightarrow_p (r_e, r_f)$, needs $\frac{1}{\text{scv}^{\beta_T}} + \frac{\text{scv}^{\beta_T}}{T^{1-\alpha/2}} \rightarrow 0$ (PSY, eq. 11)
- The critical value must diverge, so that the false-alarm probability vanishes ($\beta_T \rightarrow 0$), but slower than $T^{1-\alpha/2}$, so that true bubbles are still found; e.g. growing like $\ln T$ [Phillips, Shi & Yu \(2015a\)](#)
- In practice β_T is fixed at 5% at every date: each date keeps a 5% false-alarm chance, and over hundreds of dates a false alarm becomes likely (size study below)

Appendix: The Price-Dividend Ratio: Comparison with PSY

- ▣ Our sample: 1 866 monthly levels, Jan 1871–Jun 2026; critical values from 1 000 Monte Carlo paths
- ▣ PSY, on data to 2010 and without our filter of 8 consecutive months, also date 1879–1880, 1955–1956 and 1986–1987 [Phillips, Shi & Yu \(2015a\)](#)
- ▣ With a shorter minimum window 1879–1880, 1954–1956 and 2008–2009 appear in our data too (Seminar 17, B1)

Appendix: Family-Wise Critical Values: Bootstrap Design

- ▣ Phillips and Shi: the 95% quantile of the maximum of BSADF over τ_b consecutive dates sets the FWER at 5% for the bootstrap design: τ_b monitoring dates after a start-up of w_0 observations [Phillips & Shi \(2020\)](#)
- ▣ Residuals of $\Delta y_t = a + e_t$, drawn with replacement and multiplied by $N(0, 1)$ weights; paths of $w_0 + \tau_b - 1$ changes; 999 replications
- ▣ Here $\tau_b = 104$ weeks (two years); applied to BSADF over the full history, whose later dates search more start points, the 5% control is approximate, not proved
- ▣ The pointwise rule gives a crossing within the last 104 weeks in 42% of null paths

Appendix: Real Time vs Hindsight: Definitions

- Start: first exceedance of the chain of BSADF episodes that leads to the peak, dated in hindsight
- Confirmed: after L consecutive exceedances, the first date the episode is known in real time, but with w_0 and L set from the full-sample T (a truly real-time design fixes them ex ante)
- Leads in months (GameStop: calendar days converted to months); rise: from confirmation to the peak; fall: to the lowest close within 2 years (GameStop: 1 year)
- Bitcoin 2021: the explosive chain peaked at the local high of 13 Apr 2021 and ended in May 2021; the all-time high of 8 Nov 2021 (episodes table) came after a deep fall and a slower recovery that BSADF did not flag

Appendix: Real-Time Monitoring: Newer Procedures

- Repeating a one-shot test every period inflates the false-positive rate; Astill et al. compare monitoring statistics with the maximum over a training period, so the false-positive rate is set for a chosen monitoring horizon, robust to time-varying volatility [Astill et al. \(2018\)](#)
- Bubbles and crashes: after a bubble is detected, monitor for the crash with the sign of the mean of Δy_t (explosive vs stationary regime); never signals a crash before it starts [Whitehouse, Harvey & Leybourne \(2023\)](#)
- Earlier detection: alternative variance standardisations keep the false-detection probability of existing procedures but find bubbles sooner (applied to Bitcoin and housing) [Whitehouse, Harvey & Leybourne \(2025\)](#)
- Collapse phase: PSY applied to the reversed sample dates the crisis and the recovery [Phillips & Shi \(2018\)](#)
- Alternatives to sup-ADF: a Chow-type break test from random walk to explosive has the highest power in Homm and Breitung [Homm & Breitung \(2012\)](#); recursive tests with a start-point search [Harvey, Leybourne & Sollis \(2015\)](#)

Appendix: LPPL in the Cryptocurrency and Romanian Markets

- Bucharest Stock Exchange: herding and log-periodic patterns before the 2007–2008 crash [Pele, Mazurencu & Nijkamp \(2013\)](#); crash modelling for the BVB [Pele & Mazurencu-Marinescu \(2012\)](#)
- Cryptocurrencies: a fundamental value from Metcalfe's law (network size) combined with LPPL bubbles [Pele & Mazurencu-Marinescu-Pele \(2019\)](#), [Conda et al. \(2026\)](#)
- Dogecoin 2021: LPPL together with sentiment from social media [Pele \(2025\)](#)
- Common lesson: LPPL describes the acceleration well; the crash date remains the weak point

Appendix: Estimation: Hamilton Filter, Kim Smoother, EM

- Log-likelihood by prediction and update: $\ell(\theta) = \sum_t \ln \sum_j P(s_t = j \mid \mathcal{F}_{t-1}) \varphi(r_t; \mu_j, \sigma_j^2)$, $\mathcal{F}_t$ = returns up to t , $\varphi(x; \mu, \sigma^2)$ the Normal density with mean μ and variance σ^2 [Hamilton \(1989\)](#)
- Kim smoother, backwards: $P(s_t = j \mid \mathcal{F}_T) = P(s_t = j \mid \mathcal{F}_t) \sum_k p_{jk} \frac{P(s_{t+1} = k \mid \mathcal{F}_T)}{P(s_{t+1} = k \mid \mathcal{F}_t)}$ [Kim \(1994\)](#)
- EM algorithm: with smoothed $\xi_{tj} = P(s_t = j \mid \mathcal{F}_T)$ and $\xi_{t,ij} = P(s_{t-1} = i, s_t = j \mid \mathcal{F}_T)$, the M-step is weighted least squares [Hamilton \(1990\)](#)
 - ▶ $\hat{\mu}_j = \frac{\sum_t \xi_{tj} r_t}{\sum_t \xi_{tj}}$, $\hat{\sigma}_j^2 = \frac{\sum_t \xi_{tj} (r_t - \hat{\mu}_j)^2}{\sum_t \xi_{tj}}$, $\hat{p}_{ij} = \frac{\sum_{t \geq 2} \xi_{t,ij}}{\sum_{t \geq 2} \xi_{t-1,i}}$
- The likelihood is unbounded (one regime on a single return with $\sigma_j \rightarrow 0$) and has several local maxima (regime labels can switch): we take the best interior maximum from 20 random starts and check that no $\hat{\sigma}_j$ collapses towards 0, also in bootstrap refits
- Standard errors from the inverse numerical Hessian; they assume the number of regimes is known

Appendix: Why the LR Test for Regimes Is Non-Standard

- Under H_0 the regime means and variances coincide, so p_{11}, p_{22} are not identified (nuisance parameters present only under H_1); the information matrix is singular and some scores are identically zero [Hansen \(1992\)](#)
- Hansen: bounds on the standardised LR via empirical-process theory [Hansen \(1992\)](#)
- Carrasco, Hu and Ploberger: an optimal test needing only the estimate under H_0 , with bootstrap critical values [Carrasco, Hu & Ploberger \(2014\)](#)
- Same logic for two vs three regimes: the null model is then the fitted two-regime chain

Appendix: Case Study, Data and Method

- Data (Section 1.1): 51 580 408 arm's-length home sales from deeds records, 1995–2014, in 115 metropolitan areas (48% of the US housing stock); the cycle analysis covers 2000–2011, since realised holding periods need consecutive sales
 - ▶ Listings of homes for sale linked to the sales at property level in 57 of these areas
- Speculators (Section 1.2): short-term buyers resell within three years (realised holding period); non-occupant buyers give a mailing address different from the property address (102 areas)
- Lead-lag (Eq. 1, Figure 2): $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$ for $k = -12, \dots, 48$ months
 - ▶ p : log price, demeaned by city; v : sales divided by the 2000 housing stock, demeaned by city and calendar month; implied correlation $\beta_k \text{sd}(v_{i,t-k}) / \text{sd}(p_{i,t})$, standard errors clustered by month
- Cross-section (Table 2): OLS across cities of the price boom (2000–2006), price bust (2006–2011), listings, volume and foreclosures on the speculative volume boom (2000–2005)
- Model (Sections 5–6): buyers extrapolate recent price growth and differ in horizon; sellers post list prices and not every listing sells (non-Walrasian market)
 - ▶ Counterfactuals: rational expectations, no speculators, transaction taxes

Appendix: Case Study, Complete Results

- Figure 2: the correlation between prices and lagged volume peaks at a lag of 24 months; Figure 4: sales rise from 2.73 to 3.82 million (2000 to 2005), short-term sales from 510 to 940 thousand, 39% of the rise (non-occupant purchases: 53%)
 - ▶ Interpretation: volume leads prices by about two years, and speculators account for much of the volume boom
- Table 2, Panel A: coefficient of the short-term sales volume boom in the price-boom regression 1.930 (standard error 0.297, $R^2 = 0.272$); one standard deviation more (12.9%, Table 1) means a price rise larger by 24.9 percentage points; price bust -0.571 (0.083), 7.4 points deeper
 - ▶ Interpretation: cities with more speculation had both larger booms and larger busts
- Table 2, Panels B–C (quantities in % of 2003 total volume): unsold listings in the quiet 5.961 (1.353), 76.9 points more per standard deviation; foreclosures 0.895 (0.398), 11.5 points, about 370 thousand more across the 115 areas (Section 3.2)
 - ▶ Interpretation: consistent with speculators who bought in the boom listing in the quiet and failing to sell; that this turns the cycle is a model result (Sections 5–6), not identified by the regressions
- Figure 7, Panel C: a 10% price gain over the past year goes with an 8.2 points larger short-term buyer share; Table 6: in the model, a 5% purchase tax on all buyers cuts the price bust from 8.2% to 1.1%, on non-occupants only to 5.8%
 - ▶ Interpretation: extrapolation draws short-horizon buyers in; short horizons, not non-occupancy, amplify the cycle

Appendix: Case Study, Lead–Lag Inference

- p and v demeaned over the full sample; implied correlation $\hat{\beta}_k \text{sd}(v_{t-k})/\text{sd}(p_t)$, $k = -12, \dots, 48$ months
- At $k = 0$ the correlation is 0.03, at $k = -12$ it is -0.34 : level correlations, not a directional test of whether prices lead volume
- Eq. (1) has no constant and the lag-aligned series need not have zero mean: the “implied correlation” is a standardised slope, not a Pearson correlation, so bootstrap limits above 1 are possible
- The block bootstrap assumes stable, weakly dependent series, doubtful for one boom–bust cycle in levels: the band is pointwise and exploratory
- 1990–2011: the maximum moves to the edge of the grid, $k = 48$ (0.82)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** map what is known, then verify every reference
 - ▶ “Which papers control the false-alarm rate of sequential bubble monitoring, and what size and power do they report? Give DOI and simulation design.”
 - ▶ Resolve each DOI on Crossref and find the simulation table the answer quotes; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing explanations, each with its own observable
 - ▶ “For the current NVIDIA episode, list competing explanations (rational bubble, explosive expected cash flows, falling discount rate) and the observable each one predicts.”
 - ▶ E.g. a bubble predicts an explosive price-to-earnings ratio; explosive cash flows predict explosive earnings with a stable ratio
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Implement GSADF with $r_0 = 0.01 + 1.8/\sqrt{T}$ on weekly Bitcoin closes from the course data and simulate its 95% critical value.”
 - ▶ Replicate first: GSADF = 3.85 and critical value 2.27 on 626 weeks, as in this chapter

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the alarms are evaluated
 - ▶ “Pre-register the evaluation of the alarm: horizons 30/60/90/180 days, crash thresholds 10/20/30%, inference robust to overlapping windows, Holm correction over the 12 cells.”
 - ▶ Twelve cells tried and one reported is data snooping; twelve cells pre-registered and corrected is a test
- **Critique**: ask the assistant to attack the result
 - ▶ “Act as a referee: why could an alarm-crash association be an artefact of overlapping windows, look-ahead or the choice of assets?”
 - ▶ Expect: dependent observations inflate significance; full-sample critical values; famous episodes only, the selection that Greenwood, Shleifer & You (2019) avoid
- **Report**: hit rates with valid p -values, all 12 cells, the alarms that failed
 - ▶ The prompts that shaped the design go into AI_USE.md

Appendix: What the Researcher Must Check

- **Full-sample information in date-stamping:** critical values, lag order or detrending fitted on the whole sample
 - ▶ Test: cut the sample at t and recompute; the flag at t must not change
- **Wrong tail:** left-tail ADF critical values (about -2.86) used against an explosive alternative
 - ▶ Test: on simulated random walks the right-tailed test must reject in about 5% of the paths
- **Changing volatility:** Monte Carlo critical values under constant variance distort the size
 - ▶ Test: wild bootstrap critical values; for Bitcoin 2.82 instead of 2.27
- **Multiplicity:** many dates, assets and windows; some alarm is almost certain
 - ▶ Test: run the monitoring rule on simulated random walks and count the paths with at least one alarm
- **Overlapping outcomes:** 90-day windows overlap, so Fisher's exact test or a χ^2 test overstates significance; windows cut by the end of the data must be dropped
 - ▶ Test: block bootstrap or circular rotations of the alarm series, which keep the dependence of each series; rotations also need stationarity (mini-case)

Appendix: Mini-Case: What the Result Does and Does Not Say

- A large one-sided p is not evidence of **no** information: it only fails to support a **higher** crash probability after alarms
 - ▶ Both point estimates are **negative**: rotation p for a lower crash rate after alarms 0.07 (Bitcoin), 0.03 (Nasdaq 100); block-bootstrap CI and Newey–West test in Seminar 17, B6
 - ▶ A post-hoc, two-sided reading: alarms come in steady rises that did not end in a crash within 90 days, not an announced crash
- Assumptions behind the rotation p -values: under H_0 every circular shift of the alarm series (at least 26 steps, both ends joined) is as likely as the observed alignment
 - ▶ This is stronger than equal hit rates: it needs stationary series and alarms independent of the crash process; shared regimes or a time-varying crash rate can invalidate it
 - ▶ Hence the p -values are exploratory; their size should be checked by simulation under plausible null processes
- Outcomes: only dates whose whole horizon is observed (no censored windows at the end of the data)

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Cryptocurrency: Speculative Asset and Medium of Exchange
- ☐  Are Cryptos becoming alternative Assets?
- ☐  Statistics of Financial Markets

Code and data

- ☐ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_17

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